MULTI-TASK PREDICTION OF SOLID-PHASE EXTRACTION PROTOCOL COMPONENTS FROM MOLECULAR DESCRIPTORS

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ABSTRACT

Solid-phase extraction (SPE) is widely used to enrich trace pollutants before instrumental analysis, yet selecting cartridge chemistries, solvents and mixture ratios still relies on literature search and expert adaptation. Learning these choices from published methods is difficult because a pollutant can have several reported alternatives, solvent labels are highly imbalanced, and ratio annotations are incomplete. We propose SPEGen (Solid-Phase Extraction Protocol Generator), a multi-task framework that predicts SPE components from molecular descriptors. We aggregate published methods by pollutant to construct set-valued supervision that retains alternative reported choices. A shared encoder with task-specific heads predicts cartridge sets, workflow-step indicators, step-specific solvent sets and their ratios, with ratio training restricted to observed annotations. Rank-Gauss normalisation handles heavy-tailed descriptors, while a validation-tuned decoder selects label sets and suppresses solvents for absent steps. Bootstrap aggregation combines predictions from resampled training data. Experiments against six tabular baselines on a held-out pollutant split show that SPEGen-Bag achieves the highest cartridge and solvent F1 and the lowest ratio mean absolute error. Cumulative ablations show gains from normalisation and bagging, while decoding analyses reveal how predicted set size affects the trade-off between recovering reported alternatives and exact-match accuracy.

1 INTRODUCTION

Detecting trace pollutants in environmental samples requires enriching the analytes and reducing matrix interference before instrumental analysis. Solid-phase extraction (SPE) is widely used for this purpose because sorbents and solvents can be adapted to different compounds and sample matrices (Hennion, 1999; Poole, 2003). Developing an SPE method nevertheless requires several related decisions: which cartridge chemistry to use, which workflow steps to include, and which solvents and mixture ratios to assign to each step. These choices depend on analyte properties and sorbent–solvent interactions (Fontanals et al., 2007; Andrade-Eiroa et al., 2016). Analysts typically search for published procedures for related compounds and adapt them through expert judgement (Buszewski & Szultka, 2012). This motivates learning to recommend SPE components systematically from the properties of a target pollutant.

Molecular machine learning has made substantial progress in property prediction (Cherkasov et al., 2014; Wu et al., 2018; Yang et al., 2019), including chromatographic retention-time prediction (Domingo-Almenara et al., 2019; Bouwmeester et al., 2019). These models estimate properties of a molecule, but do not directly specify the components needed for sample preparation. Related work in synthetic chemistry moves closer to experimental decisions: Gao et al. (2018) predict reaction conditions, and Vaucher et al. (2020) and Vaucher et al. (2021) extract or predict machine-readable procedures. For SPE, the learning target must accommodate both categorical selections and numeric ratios, while allowing multiple reported choices for the same pollutant.

To address this problem, we propose SPEGen (Solid-Phase Extraction Protocol Generator), which predicts SPE components from molecular descriptors. Rather than selecting one published method as the sole reference, we aggregate cartridge, step and solvent labels by pollutant. This set-valued

supervision retains alternatives across source methods. Reported ratios provide conditional numeric targets, and missing ratios are masked during training. The resulting formulation lets the model learn from partially annotated methods without assigning an arbitrary single answer to a pollutant.

The predictor uses a shared encoder with task-specific heads for cartridges, workflow steps, step-wise solvents and ratios. Sharing the molecular representation allows the tasks to use common chemical information while retaining their distinct output types. To address sparse, imbalanced labels, cartridge and solvent prediction use focal and asymmetric losses, respectively. A validation-tuned decoder then selects label sets using label-specific thresholds and gates solvent outputs by predicted step presence. This separates learning candidate scores from deciding which candidates to return.

Input representation is equally important. We use interpretable descriptors of properties such as lipophilicity and polarity, which are relevant to SPE selection, but their strongly skewed distributions remain heavy-tailed after standardisation. SPEGen therefore applies a training-fitted rank-Gauss transform before the shared encoder. Its bagged variant averages predictions from independently resampled training sets before tuning the decoder. These choices address input irregularity and sensitivity to the training sample without changing the descriptor information available to the models.

Our contributions are threefold. First, we formulate SPE component recommendation as a pollutant-level, multi-task learning problem with set-valued and partially observed targets. Second, we develop SPEGen, combining shared representation learning, task-specific objectives, rank-based normalisation and validation-tuned decoding. Third, we evaluate the framework against tabular baselines and analyse its behaviour through cumulative ablations and set-size diagnostics. The experiments demonstrate improved cartridge and solvent set recovery and lower ratio error with bootstrap aggregation, and clarify the trade-offs between broad recovery, exact match and top-ranked selection.

2 Related Work

2.1 SPE method development and analytical machine learning

SPE reviews describe sorbent selection and the roles of polarity, hydrophobicity and ionisation (Hennion, 1999; Poole, 2003; Fontanals et al., 2007; Andrade-Eiroa et al., 2016; Buszewski & Szultka, 2012). Quantitative structure–property models (Cherkasov et al., 2014) and learned molecular representations (Wu et al., 2018; Yang et al., 2019) provide complementary tools, including retention-time prediction (Domingo-Almenara et al., 2019; Bouwmeester et al., 2019). Our target is instead a collection of protocol components. Descriptor-based comparisons establish a reference setting; they do not compare representation learning alternatives.

2.2 Structured prediction of laboratory procedures

Reaction-condition prediction (Gao et al., 2018), retrosynthetic planning (Segler et al., 2018) and robotic synthesis (Coley et al., 2019) connect predictions to experimental decisions. Vaucher et al. (2020; 2021) represent procedures as machine-readable actions. The Open Reaction Database provides a schema and infrastructure for sharing reaction data (Kearnes et al., 2021). SPE component prediction differs in its heterogeneous, pollutant-aggregated targets. Evaluation on reaction data would require a separately defined input–output mapping; no cross-corpus transfer result is claimed here.

2.3 Multi-label and tabular learning

Classifier chains explicitly condition on preceding labels (Read et al., 2011), whereas a shared multi-task encoder (Caruana, 1997) shares features without defining an autoregressive joint distribution. Focal and asymmetric losses address label imbalance (Lin et al., 2017; Ridnik et al., 2021; Zhang & Zhou, 2014), and validation-based threshold selection is important for F1 (Lipton et al., 2014). Strong tree baselines remain essential on tabular inputs (Grinsztajn et al., 2022); neural alternatives include numerical embeddings and attention-based encoders (Gorishniy et al., 2022; 2021). TabPFN uses in-context prediction (Hollmann et al., 2023), and TabR retrieves training examples during prediction (Gorishniy et al., 2024). Our within-batch neighbour term is a training regulariser, not retrieval at inference, and neither model is an evaluated baseline. Finally, bootstrap aggregation (Breiman, 1996)

changes member training samples, unlike ensembles that vary only initialisation (Lakshminarayanan et al., 2017).

3 PRELIMINARY

The pollutant-level target collects four types of reported SPE components. For pollutant i, let $\mathbf{x}_i \in \mathbb{R}^{21}$ denote the molecular descriptors and define

$$\mathbf{y}_i = (\mathbf{c}_i, \mathbf{m}_i, \mathbf{S}_i, \mathbf{R}_i), \quad \mathbf{c}_i \in \{0,1\}^{42},\ \mathbf{m}_i \in \{0,1\}^{5},\ \mathbf{S}_i \in \{0,1\}^{5\times 378},\ \mathbf{R}_i \in [0,1]^{5\times 378}. \quad (1)$$

Here $\mathbf{c}_i$ indicates reported cartridge classes; $\mathbf{m}_i$ indexes loading, conditioning, washing, elution and reconstitution; $\mathbf{S}_i$ is a step-by-solvent binary indicator matrix; and $\mathbf{R}_i$ stores ratio summaries. The step order is an indexing convention, not an inferred execution sequence. The molecular table contributes pollutant identifiers, canonical SMILES, 21 PubChem-derived descriptors (Kim et al., 2023) and 42 cartridge indicators. A second table contributes 36,776 method records, including step-wise solvent strings and 26,204 records with at least one parsed ratio.

Selecting just one method would penalise other reported alternatives under component-wise scoring. We instead aggregate methods linked to the same pollutant, at the cost of losing their joint assignments. Let $\mathcal{M}_i$ be the linked methods, $\mathcal{S}(\mu, t)$ the solvent tokens reported by method μ at step t, and $\mathcal{M}_{i,t,k}$ the methods that report a ratio for step t and solvent k. For annotated pollutants, the aggregate labels are

$$\begin{aligned} m_{i,t} &= \max_{\mu\in\mathcal{M}_i} \mathbb{1}[|\mathcal{S}(\mu,t)| > 0], \\ s_{i,t,k} &= \max_{\mu\in\mathcal{M}_i} \mathbb{1}[k \in \mathcal{S}(\mu,t)], \\ r_{i,t,k} &= \frac{1}{|\mathcal{M}_{i,t,k}|} \sum_{\mu\in\mathcal{M}_{i,t,k}} r_{\mu,t,k}. \end{aligned} \quad (2)$$

The ratio mean is defined only when $|\mathcal{M}_{i,t,k}| > 0$; otherwise the cell is masked, not set to a zero-ratio target. Unions preserve marginal alternatives, not their co-occurrence. Likewise, averaging conflicting ratios can yield an unreported value; no evidence here establishes that such conflicts are negligible. Figure 1(a) illustrates elution alternatives for one pollutant; Figure 1(b) shows the resulting supervision tensors. The complete five-step example and solvent-label distribution appear in Appendix A. Aggregation produces 11,441 pollutants; 10,335 have protocol annotations and 1,106 have cartridge labels only. All models use the same random 80/10/10 split of 9,152/1,144/1,145 pollutants, fixed with seed 42.

4 METHODOLOGY

Figure 2 summarises SPEGen from descriptor preprocessing to structured component outputs. Aggregated labels and ratio masks supervise training; only molecular descriptors are required for prediction. The framework shares one molecular representation across four core tasks and supports separate bagging and exact-match variants.

4.1 RANK-GAUSS INPUT

Standardisation leaves the descriptor marginals highly skewed and heavy-tailed. We instead fit a rank-Gauss map independently to each dimension on the training split. With midpoint ranks for ties and Φ denoting the standard normal cumulative distribution function,

$$\widetilde{x}_{ij} = \Phi^{-1}\left(\frac{\mathrm{rank}_j(x_{ij}) - \frac{1}{2}}{n_{\mathrm{train}}}\right). \quad (3)$$

The fitted empirical map is applied to validation and test values by interpolation, so no test statistic enters preprocessing. It is monotone, keeps the input at 21 dimensions and gives every baseline and neural model the same chemical information. The interpolation and tail conventions are required for reproduction (Appendix G); Equation 3 specifies the training ranks.

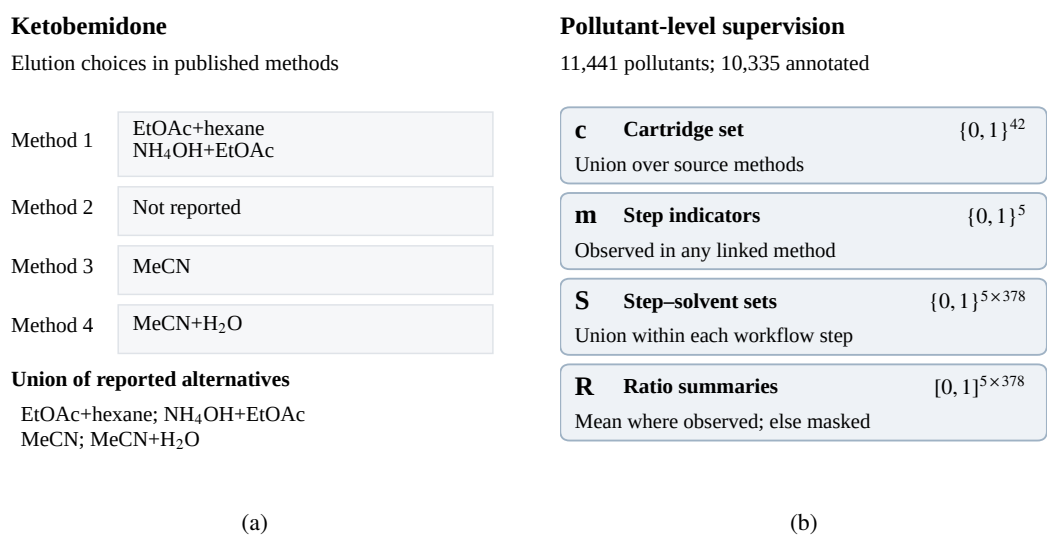


Figure 1: From published methods to set-valued supervision. (a) Elution alternatives in four literature methods for ketobemidone; an unreported step is not a negative experimental result. The full step-wise example is in Figure A2. (b) Aggregation yields cartridge, step, solvent and partially observed ratio targets.

4.2 Shared Encoder and Task Heads

In the central panels of Figure 2, parallel task heads read the shared molecular representation. A two-layer multilayer perceptron of width 512, with ReLU activations and dropout 0.3 (Srivastava et al., 2014), maps $\widetilde{\mathbf{x}}_i$ to a shared representation $\mathbf{z}_i \in \mathbb{R}^{512}$. Three linear heads produce cartridge, step and step–solvent logits. The ratio head concatenates $\mathbf{z}_i$ with learned step and solvent embeddings and predicts probabilities over 20 equal-width buckets on $[0, 1]$. The three classification heads share features, but solvent logits are not explicitly conditioned on a predicted cartridge. In compact form,

$$\begin{aligned} \mathbf{z}_i &= f_\theta(\widetilde{\mathbf{x}}_i), \\ \mathbf{u}_i^c &= W_c \mathbf{z}_i + b_c, \quad \mathbf{u}_i^m = W_m \mathbf{z}_i + b_m, \\ \mathbf{U}_i^s &= \mathrm{reshape}_{5\times 378}(W_s \mathbf{z}_i + b_s), \\ \mathbf{q}_{i,t,k} &= \mathrm{softmax}(g([\mathbf{z}_i; \mathbf{e}_t; \mathbf{f}_k])), \quad \widehat{r}_{i,t,k} = \sum_{b=1}^{20} q_{i,t,k,b} \nu_b. \end{aligned} \tag{4}$$

Here ν_b is the centre of bucket b. The ratio head has a categorical training objective and a continuous expectation decoder; neither this expectation nor target averaging preserves distinct ratio modes. Thus bucketing is a prediction parameterisation, not a guarantee of conventional mixtures. Alternative tokens may denote whole solvent systems, so ratios across their union are not mixture fractions constrained to sum to one. A bench-ready mixture requires method-level component grouping and additional validation (Appendix F).

4.3 Objective and Decoding

The training objective combines four task losses with ranking and neighbourhood regularisation,

$$\mathcal{L} = \mathcal{L}_{\mathrm{car}} + \lambda_m \mathcal{L}_{\mathrm{step}} + \lambda_s \mathcal{L}_{\mathrm{sol}} + \lambda_r \mathcal{L}_{\mathrm{ratio}} + \lambda_{\mathrm{rank}} \mathcal{L}_{\mathrm{rank}} + \lambda_{\mathrm{sm}} \mathcal{L}_{\mathrm{sm}}. \tag{5}$$

Cartridges use focal loss (Lin et al., 2017); steps use positive-weighted binary cross-entropy; solvents use the asymmetric loss of Ridnik et al. (2021); and ratios use masked cross-entropy with label smoothing over the 20 buckets. The ranking term uses reported positives; the neighbour term weights latent distances between eight within-batch descriptor neighbours. Its sign convention matters for whether it is attractive (Appendix C).

A single 0.5 threshold is inappropriate when label prevalence spans three orders of magnitude. Following the validation-based principle analysed by Lipton et al. (2014), we tune a threshold for

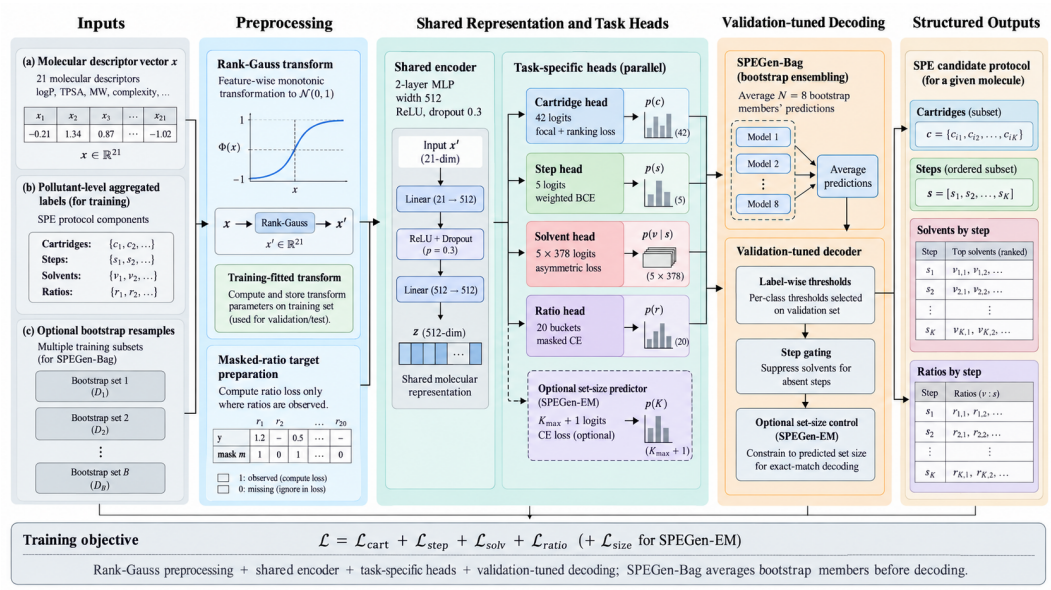


Figure 2: Overview of SPEGen. Training uses aggregated component labels and observed-ratio masks. Rank-Gauss descriptors feed a shared encoder and four task heads. SPEGen-Bag averages eight members before thresholding and step gating; SPEGen-EM uses an additional set-size predictor. The bottom band summarises task supervision; Equation 5 gives the weighted core objective with regularisation.

each sufficiently frequent label over a 41-point grid and apply a global shift to rarer labels. Decoding is

$$\begin{aligned} \widehat{c}_j &= \mathbb{1}[\sigma(u_j^c) > \tau_j^c], \qquad \widehat{m}_t = \mathbb{1}[\sigma(u_t^m) > \tau_t^m], \\ \widehat{s}_{t,k} &= \mathbb{1}[\sigma(u_{t,k}^s) > \tau_{t,k}^s]\, \widehat{m}_t. \end{aligned} \tag{6}$$

This prevents solvents under absent steps, but does not force a nonempty solvent set for every present step. We denote the balanced, chain-oriented, exact-match and bagged operating points by SPEGen-Bal, SPEGen-Chain, SPEGen-EM and SPEGen-Bag, respectively. SPEGen-Bal and SPEGen-Chain use micro-F1 threshold tuning with different weights on cartridge ranking and solvent recovery. SPEGen-EM additionally uses a set-size predictor with an EM/Jaccard decoding objective; it is an auxiliary variant, not an unchanged four-head model. SPEGen-Bag averages the probabilities of eight SPEGen-Chain members trained on independent bootstrap resamples (Breiman, 1996), then tunes thresholds once on the averaged validation predictions. Averaging before thresholding retains the probability information required by the decoder.

5 Experiments

5.1 Experiment Setting

All models receive the same 21 descriptors and the fixed random split of 9,152/1,144/1,145 training/validation/test pollutants. We compare decision tree, random forest, SVM, MLP, XGBoost and LightGBM (Breiman et al., 1984; Breiman, 2001; Chen & Guestrin, 2016; Ke et al., 2017; Pedregosa et al., 2011). We report cartridge micro-F1 (C-F1), exact match (C-EM), solvent micro-F1 (S-F1), ratio error, cartridge top-1 (C-top1), chain accuracy and ratio-verified Hit@1. SSS, HC and GUS are auxiliary summaries defined in Appendix D. Single-network rows give mean ± standard deviation over four seeds; baselines and SPEGen-Bag each have one reported evaluation. Column-best results are bold; only ratio MAE is minimised. Training and evaluation details are in Appendices C and D.

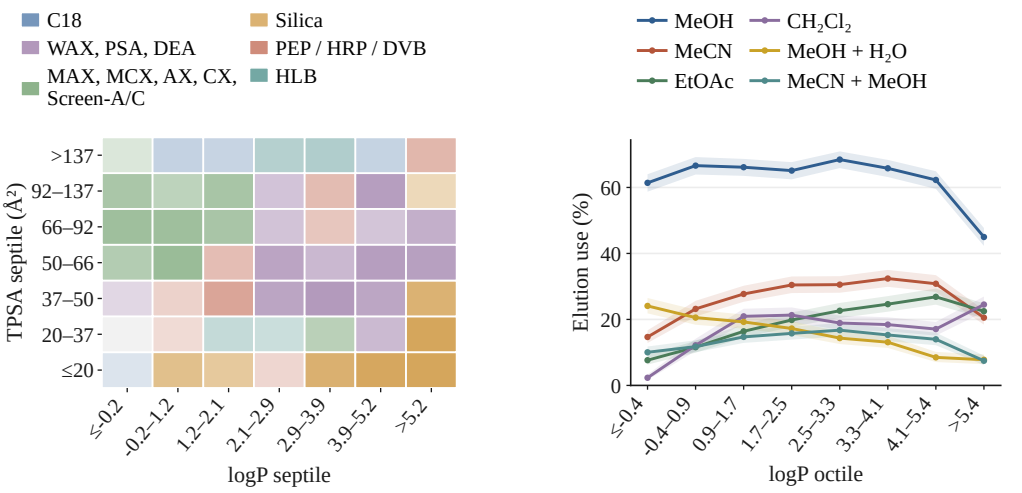


Figure 3: Chemical structure in the supervision. (a) Locally dominant cartridge classes across logP–TPSA quantile bins, with opacity proportional to enrichment over the corpus prior. (b) Prevalence of leading elution solvents across logP octiles with Wilson 95% confidence intervals. MeOH, methanol; MeCN, acetonitrile; EtOAc, ethyl acetate; CH_2Cl_2, dichloromethane.

Table 1: Per-task performance on the held-out test split ($n = 1{,}145$)

Model	C-F1	C-EM	S-F1	Ratio MAE	Ratio≤0.10
Decision tree	0.4020	0.0786	0.3933	0.1740	0.5542
Random forest	0.4030	0.1293	0.4497	0.1524	0.4898
SVM	0.0587	0.0236	0.0069	0.2174	0.3734
MLP	0.4171	0.1092	0.4567	0.1611	0.4904
XGBoost	0.4115	0.1188	0.4563	0.1850	0.3915
LightGBM	0.4181	0.1153	0.4509	0.1764	0.4115
SPEGen-Bal	0.4799 ± 0.0077	0.0155 ± 0.0018	0.4660 ± 0.0008	0.1581 ± 0.0028	0.6373 ± 0.0049
SPEGen-Chain	0.4781 ± 0.0049	0.0179 ± 0.0020	0.4538 ± 0.0026	0.1570 ± 0.0009	$\mathbf{0.6395 \pm 0.0026}$
SPEGen-EM	0.3490 ± 0.0144	$\mathbf{0.1443 \pm 0.0083}$	0.4716 ± 0.0054	0.1594 ± 0.0019	0.6341 ± 0.0033
SPEGen-Bag	**0.4909**	0.0384	**0.4747**	**0.1464**	0.5916

5.2 Chemical Signal

We first examine chemical associations in the descriptor space. Figure 3(a) shows that lipophilic, low-TPSA pollutants favour silica, whereas more polar regions favour mixed-mode ion exchangers and broader reversed-phase materials. The same descriptors also correlate with solvent choice: methanol becomes less common in the most lipophilic bins, while ethyl acetate and dichloromethane become more common. These full-corpus analyses do not enter model selection; additional enrichment maps and cartridge–solvent association plots are deferred to Appendix B.

5.3 Main Result

Tables 1 and 2 compare four operating points of SPEGen with six tabular baselines under the same split and descriptor input. The variants differ in loss weights, decoding objectives and the optional size head; SPEGen-Bag ensembles eight SPEGen-Chain members (Appendix Table A2).

On per-task prediction, SPEGen-Bag improves cartridge F1 from 0.4181 for LightGBM to 0.4909 and solvent F1 from 0.4567 for the MLP to 0.4747. Its ratio MAE is 0.1464, compared with 0.1524 for the strongest baseline. SPEGen-Chain gives the highest ratio-within-0.10 rate, 0.6395 versus 0.5542 for the decision tree. SPEGen-EM raises cartridge exact match to 0.1443, above the random forest's 0.1293, but its smaller predicted sets reduce cartridge F1 to 0.3490.

Table 2: Structured-output performance on the test split ($n = 1{,}145$ overall)

Model	C-top1	Chain acc.	Hit@1	SSS	HC	GUS
Decision tree	0.4760	0.2790	0.2464	0.4660	0.8855	55.31
Random forest	0.5939	**0.5168**	0.4468	0.4848	0.8023	56.97
SVM	0.4699	0.3471	0.2809	0.2573	0.9041	31.05
MLP	0.5642	0.4506	0.3960	0.4666	0.8309	57.39
XGBoost	0.5965	0.5091	0.3988	0.4695	0.8084	55.96
LightGBM	0.5921	0.4746	0.3787	0.4680	0.8242	56.20
SPEGen-Bal	0.5775 ± 0.0048	0.4859 ± 0.0033	0.4422 ± 0.0043	0.4810 ± 0.0008	0.9072 ± 0.0022	$\mathbf{61.66 \pm 0.15}$
SPEGen-Chain	0.5948 ± 0.0048	0.4952 ± 0.0049	$\mathbf{0.4521 \pm 0.0039}$	0.4750 ± 0.0027	$\mathbf{0.9083 \pm 0.0017}$	61.22 ± 0.23
SPEGen-EM	0.5753 ± 0.0032	0.4705 ± 0.0056	0.4243 ± 0.0043	$\mathbf{0.4914 \pm 0.0011}$	0.8989 ± 0.0059	58.52 ± 0.41
SPEGen-Bag	**0.6140**	0.4995	0.4506	0.4751	0.9070	61.55

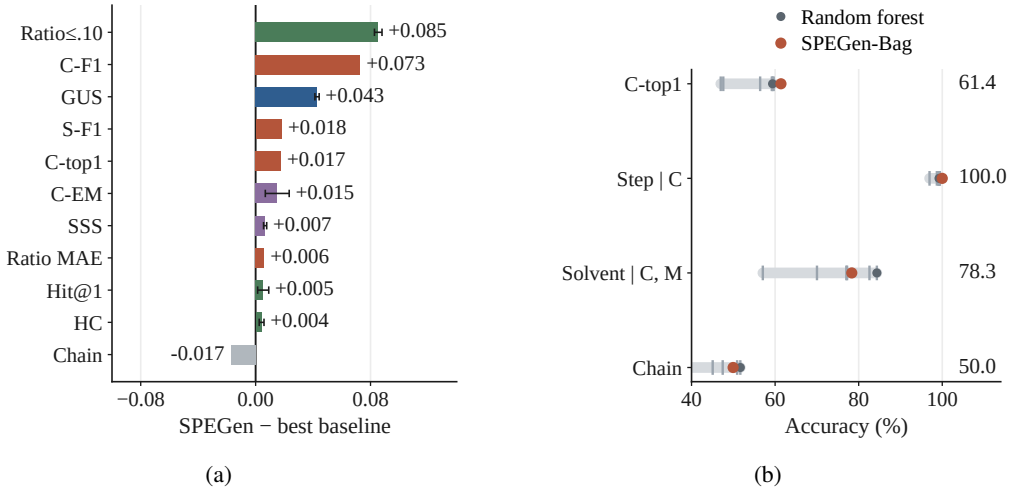
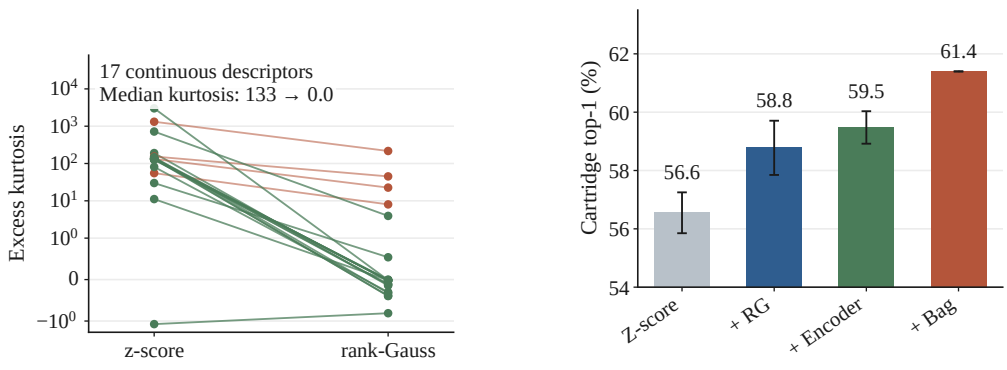


Figure 4: Held-out comparison with tabular baselines. (a) Margin between the best SPEGen variant and the strongest baseline for each metric; error bars show seed variation for single networks. GUS differences are divided by 100. (b) Conditional chain diagnostics: grey spans the baseline range, dark grey marks random forest, and orange marks SPEGen-Bag; C and M denote a correct cartridge and step.

On structured metrics, SPEGen-Bag reaches the highest cartridge top-1 accuracy (0.6140 versus 0.5965 for XGBoost), while random forest remains strongest on chain accuracy (0.5168 versus 0.4995 for SPEGen-Bag). Figure 4(a) summarises the margin between the best reported SPEGen variant and the strongest baseline on each metric, and Figure 4(b) points to solvent ranking as a remaining weakness.

The gains do not come from a single universally best configuration. SPEGen-Bal leads GUS, SPEGen-Chain leads Hit@1 and HC, and SPEGen-EM leads cartridge exact match and SSS. Across variants, the best reported point estimates exceed the baselines on ten metrics, but this is not one model's score profile or a significance test. The single-network SPEGen-Chain reaches cartridge top-1 of 0.5948, below XGBoost's 0.5965; the 0.6140 result depends on ensembling. The ratio results also distinguish average error from tolerance accuracy: SPEGen-Bag minimises MAE, whereas SPEGen-Chain places more predictions within 0.10 of the target.

For solvent top-1 given a correct prefix, the conditional diagnostic is 0.783 for SPEGen-Bag versus 0.844 for random forest, while step-given-cartridge is 1.000 versus 0.992. This suggests that ranking a single solvent remains harder than recovering a solvent set. The conditional scores require matched evaluation subsets before multiplication into a chain score, and the comparison is not compute-matched (Appendix D).



(a) (b)

Figure 5: Cumulative ablation. (a) Descriptor excess kurtosis under z-scoring and rank-Gauss normalisation; four near-constant counters remain resistant to a monotone transform. (b) Cumulative cartridge top-1 accuracy after rank-Gauss (RG) normalisation, encoder tuning and bootstrap bagging. Error bars apply to the four-seed single-network stages; the ensemble has one reported evaluation.

Table 3: Stage-wise ablation of SPEGen

Configuration	C-top1	Chain	Hit@1	C-F1	S-F1	GUS	n
Z-score, default encoder	0.5655 ± 0.0070	0.4626 ± 0.0106	0.4130 ± 0.0119	0.4574 ± 0.0074	0.4356 ± 0.0035	59.91 ± 0.29	4
+ rank-Gauss	0.5878 ± 0.0093	0.4851 ± 0.0095	0.4389 ± 0.0095	0.4658 ± 0.0062	0.4420 ± 0.0016	60.42 ± 0.17	4
+ tuned encoder	0.5948 ± 0.0056	0.4952 ± 0.0057	$\mathbf{0.4521 \pm 0.0045}$	0.4781 ± 0.0057	0.4538 ± 0.0030	61.22 ± 0.26	4
+ bootstrap bagging	**0.6140**	**0.4995**	0.4506	**0.4909**	**0.4747**	**61.55**	1

5.4 Ablation

Table 3 adds rank-Gauss normalisation, encoder tuning and bootstrap aggregation sequentially; n counts evaluations, not ensemble members. Rank-Gauss raises cartridge top-1 from 0.5655 ± 0.0070 to 0.5878 ± 0.0093, while also improving all other reported metrics. Figure 5(a) shows the accompanying distributional change: under z-scoring, many descriptor dimensions remain strongly heavy-tailed, whereas rank-Gauss reduces the median excess kurtosis of the 17 continuous dimensions from 133 to approximately zero.

Encoder tuning adds a further 0.0070 in cartridge top-1, and eight-member bootstrap aggregation adds 0.0192 more. Bagging also produces the best cartridge F1 and solvent F1, although Hit@1 changes from 0.4521 to 0.4506, showing that broader set recovery need not improve ratio-verified top-1 selection. These increments are conditional on the order of the cumulative ablation, rather than isolated causal effects. Additional single-run interventions, including numerical embeddings and uncertainty weighting (Gorishniy et al., 2022; Kendall et al., 2018), are summarised in Appendix E.

5.5 Decoding Analysis

The SPEGen variants exhibit a precision–recall trade-off. F1-oriented decoding predicts 6.2–6.5 cartridge labels against 3.3 reported labels, obtaining recall near 0.69 at precision near 0.38. Exact-match decoding predicts 1.4 labels, raises precision to 0.58 and obtains the best cartridge exact match in the study (0.1443), but reduces cartridge F1 to 0.3490.

For one diagnostic run, rather than the four-seed main-table means, a fixed-ranking oracle-size intervention raises cartridge F1 from 0.4805 to 0.6532 and exact match from 0.0183 to 0.2498; solvent F1 rises from 0.4564 to 0.6470 (Appendix Table A4). A dedicated gradient-boosted count predictor obtains test R^2 values of 0.09 and 0.05 and Spearman correlations of 0.31 and 0.32 for cartridge and solvent counts, respectively. These are weak results for this count predictor, not a bound on all descriptor-based predictors. The precision–recall and cardinality plots are in Figure A9; fixed-ranking oracle conventions are detailed in Appendix E.

5.6 INTERPRETABILITY

A depth-three CART model (Breiman et al., 1984) on 2,924 single-cartridge pollutants reaches held-out balanced accuracy 0.31, against chance 0.17. Figure 6 shows silica enrichment of $5.2\times$ for complexity above 374 and logP above 6.75. The solvent tree reaches 0.30 and is retained in Appendix Figure A7. These restricted-subset trees describe chemical associations; they are not surrogates of SPEGen and their balanced accuracies are not directly comparable with its multi-label scores.

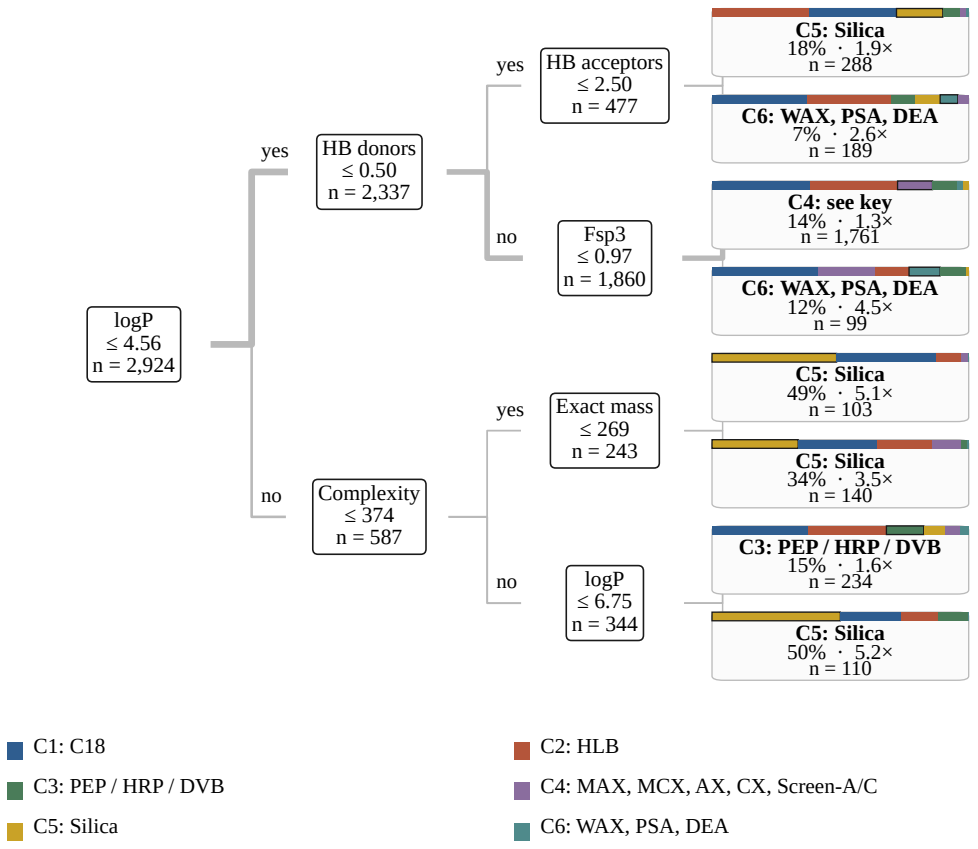


Figure 6: Descriptor-level cartridge rules on the single-cartridge subset. Leaves show the predicted class, observed share, enrichment and sample count; stacked bars show class composition, with the predicted class outlined. Edge width reflects sample flow. Class-balanced fitting need not select the largest unweighted leaf segment.

6 CONCLUSION

SPEGen learns SPE component recommendations from pollutant-level aggregation of published methods. On the fixed held-out split, its bagged variant improves cartridge and solvent set recovery and achieves the lowest ratio MAE among the reported methods. Cumulative ablations associate gains with rank-Gauss normalisation and bootstrap aggregation, while decoding analyses expose the trade-off between broad recovery and exact match. Descriptor-level trees provide complementary evidence of chemical associations in the corpus. The outputs remain candidate components: incomplete reporting and loss of source-method co-occurrence require further compatibility checks and prospective laboratory validation.

AI USE STATEMENT

Generative AI tools assisted the design and refinement of the research methodology and experiments in this work. We also used AI assistance to create and revise scientific figures and their plotting code, and to edit the manuscript for clarity and readability. AI-assisted figure revisions concern the presentation of the study's data and experiment outputs, not additional experimental evidence. AI suggestions do not replace scientific verification. The authors retain responsibility for the research decisions, the accuracy and originality of the submission, and all text, code and visual material included in it.

ETHICS STATEMENT

We acknowledge the ICLR Code of Ethics and the responsibility to conduct and communicate this research transparently. This retrospective computational study uses literature-derived SPE records and molecular descriptors to support analytical method development. Uneven literature coverage and positive-only reporting can bias the recommendations; an unreported choice is not a demonstrated chemical failure. Moreover, pollutant-level aggregation does not preserve a jointly validated procedure. The outputs are candidate components, not instructions for autonomous laboratory execution. Any experimental application requires expert assessment of chemical compatibility, solvent hazards and waste handling, followed by laboratory validation. Use and redistribution of the source records and derived resources must respect the original sources' rights and access conditions. The implications of aggregation and the limits of generalisation are discussed in Appendices F and G.

REPRODUCIBILITY STATEMENT

All source code for data processing, model construction and training is included in the supplementary attachment, together with the decoding, evaluation, baseline and figure-generation routines. Sections 3 and 4 describe the task, model and decoder, and Section 5.1 specifies the split and experimental protocol. Appendix A documents target construction and missing annotations; Appendix C and Table A2 describe the objectives and training settings; Appendix D discusses evaluation conventions; and Appendix E presents additional diagnostics. The attached experiment scripts, saved configurations and outputs, and repository documentation provide the implementation record and figure-rebuild instructions. These materials support inspection and rerunning of the pipeline; remaining run-provenance and reporting limitations are identified in Appendix G.

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APPENDIX

A CORPUS CONSTRUCTION AND SUPERVISION STATISTICS

Table A1 consolidates the target construction, learning signals and decoding rules. The protocol parser processes five steps—loading, conditioning, washing, elution and reconstitution—and converts every non-empty cell into one or more normalised solvent tokens. A numeric ratio, when present, is associated with its step–solvent pair. Cells without a ratio still supervise solvent classification but are masked from ratio prediction. Pollutant-level aggregation takes unions over cartridge, step and solvent labels and averages repeated ratios, as defined in Equation 2.

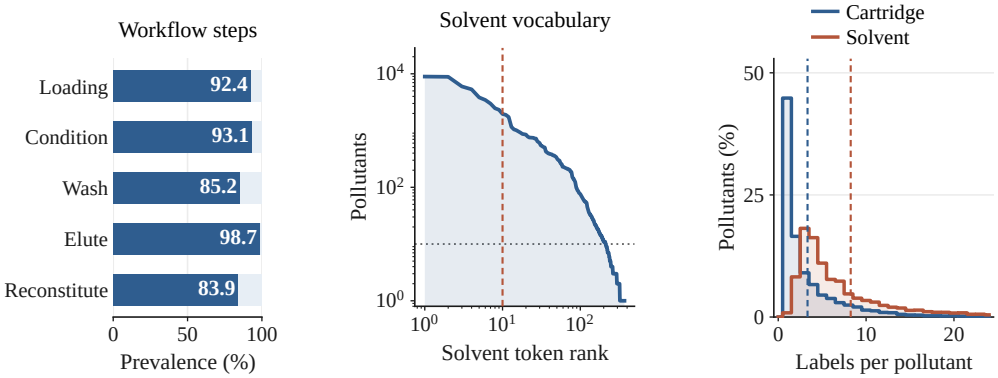


Figure A1: Shape of the supervision signal. (a) Workflow-step prevalence among annotated pollutants. (b) Solvent-token rank frequency. (c) Distributions of reported cartridge and solvent set size; dashed lines mark their means.

Figure A1 characterises step prevalence, solvent frequency and reported set sizes. Elution appears in 98.7% of annotated pollutants and reconstitution in 83.9%, so the step mask contains relatively little discriminative information. The solvent vocabulary is strongly long-tailed: the ten most frequent tokens account for 54% of all positive uses and 157 of 378 tokens occur fewer than ten times. Cartridge and solvent set sizes are right-skewed, with means 3.34 and 8.25. Numeric ratios are observed for 39.7% of positive step–solvent cells. Descriptor kurtosis before and after rank-Gauss normalisation is shown in Figure 5(a).

Table A1: Structured label space, learning signals and decoding

Component	Symbol	Size	Construction	Learning and decoding
Molecular descriptors	$\mathbf{x}_i$	21	PubChem descriptors from Complexity to logP; rank-Gauss normalised on the training split	Shared encoder input
Cartridge set	$\mathbf{c}_i$	42	Union of cartridge classes reported for the pollutant; 3.34 labels on average	Focal loss plus ranking; per-label thresholds
Step mask	$\mathbf{m}_i$	5	Union over loading, condition, wash, elute and reconstitute	Positive-weighted BCE; per-step thresholds
Step-wise solvents	$\mathbf{S}_i$	5×378	Union of normalised tokens conditioned on the step; 8.25 distinct solvents per protocol	Asymmetric loss; per-label thresholds gated by steps
Mixture ratios	$\mathbf{R}_i$	5×378	Values in $[0, 1]$; mean of reported ratios; missing cells are masked	Masked 20-bucket cross-entropy; bucket expectation
Corpus	–	11,441	10,335 pollutants with protocol annotation and 1,106 with cartridge labels only	–

B CHEMICAL ANALYSES

For each cartridge class, prevalence within a descriptor quantile bin is divided by prevalence in the full corpus and reported on a base-2 logarithmic scale. Figure A3 shows the one-dimensional enrichment patterns; Figure 3(a) gives the joint logP–TPSA map. Silica is enriched 2.6-fold in the highest

Published methods for Ketobemidone

4 independent literature protocols, one pollutant

	Load	Condition	Wash	Elute	Reconstitute
method 1	MeOH+phosphate	MeOH H_2O phosphate	phosphate MeOH	EtOAc+hexane NH_4OH+EtOAc	MeCN+HCOOH
method 2	K-phosphate	—	—	—	NH_4HCO_2+MeOH
method 3	NH_4OAc+MeOH	MeOH H_2O	MeOH+H_2O	MeCN	NH_4OAc+MeOH
method 4	HCOOH+H_2O	MeOH H_2O	H_2O MeOH+H_2O MeOH	MeCN+H_2O	MeOH+H_2O
	▼ Union over methods; ratios averaged				
target	MeOH+phosphate K-phosphate NH_4OAc+MeOH HCOOH+H_2O	MeOH H_2O phosphate	phosphate MeOH MeOH+H_2O H_2O	EtOAc+hexane NH_4OH+EtOAc MeCN MeCN+H_2O	MeCN+HCOOH NH_4HCO_2+MeOH NH_4OAc+MeOH MeOH+H_2O

Published alternatives are retained as component-wise targets; joint method assignments are not retained.

Figure A2: Complete step-wise example for ketobemidone. Each row retains the solvents reported in one source method. The bottom row takes a union separately within each step; missing entries do not imply unsuccessful extraction. Repeated ratios, where available, are averaged.

logP octile and depleted 2.4-fold in the lowest; Florisil and aminopropyl follow the same direction. Ion-exchange materials favour the polar, high-TPSA end, whereas C18 and HLB are relatively flat. Quantile bins contain equal numbers of molecules, cells with fewer than 40 observations are blank, and aggregate "Unknown" and "Null cartridge" classes are excluded.

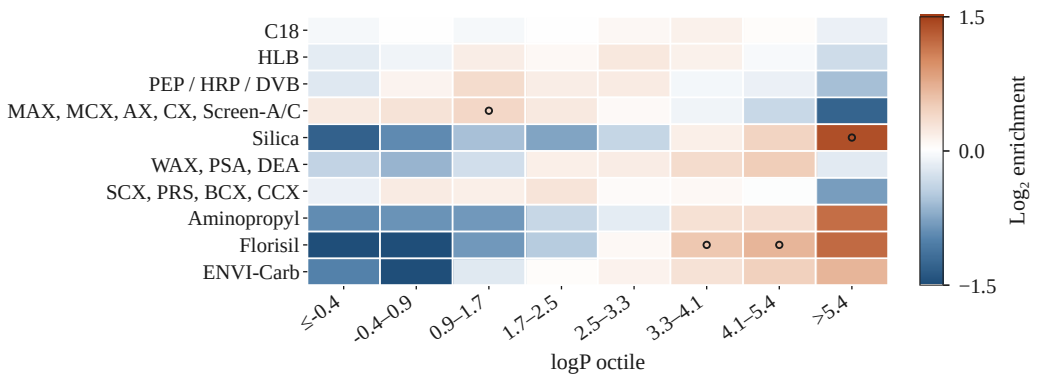
Recorded solvent choice is associated with the workflow step, the analyte and the cartridge (Figures 3(b) and A4–A6). Conditioning is dominated by methanol and water, washing is predominantly aqueous and elution is predominantly organic. Across logP octiles, methanol decreases from roughly 66% in the middle of the range to 45% in the most lipophilic octile, while hexane rises from 2% to 30%, ethyl acetate from 8% to 23% and dichloromethane from 2% to 25%. Standardised mean differences give the same ordering: hexane-eluted pollutants are 0.8 pooled standard deviations higher in logP than the remainder, whereas water-eluted pollutants are 0.6 lower. The cartridge–solvent panel is centred by row because a pollutant may carry several cartridge labels, making raw conditional probabilities difficult to interpret.

Depth-three CART models (Breiman et al., 1984) provide interpretable descriptor rules on unambiguous subsets: 2,924 pollutants with one cartridge and 2,868 protocols with one elution solvent. Held-out balanced accuracy is 0.31 for cartridge and 0.30 for solvent, compared with chance 0.17. The cartridge tree is reported in the main text (Figure 6); the solvent tree is retained below. In the latter, complexity below 144 and monoisotopic weight below 150 select dichloromethane at 6.8 times its corpus rate. Together they show that the descriptor signal is chemically coherent but incomplete, motivating a model that can use descriptor interactions and correlations among outputs.

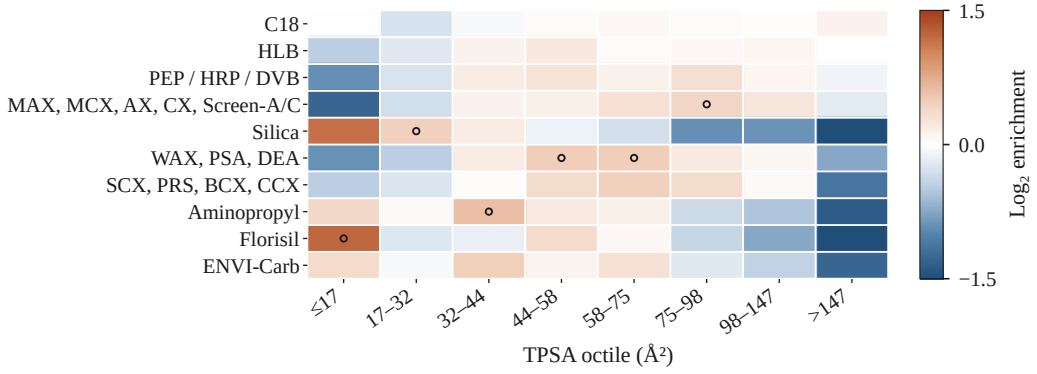
C Detailed objective and optimisation

Task losses. Writing p_j for a cartridge sigmoid probability, cartridge classification uses focal loss with $\gamma = 1.5$ (Lin et al., 2017),

$$\mathcal{L}_{\text{car}} = -\frac{1}{42}\sum_{j=1}^{42}\left[c_j(1-p_j)^{\gamma}\log p_j + (1-c_j)p_j^{\gamma}\log(1-p_j)\right]. \tag{A1}$$



(a)



(b)

Figure A3: Cartridge selection across lipophilicity and polarity. (a,b) Enrichment across logP and TPSA octiles, expressed as $\log_2[P(\text{cartridge} \mid \text{bin})/P(\text{cartridge})]$. Open circles mark the most enriched class in a bin when enrichment exceeds $1.3\times$.

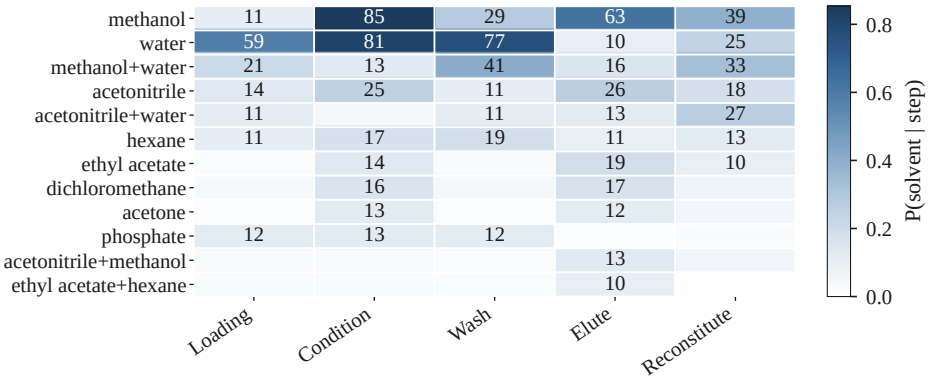


Figure A4: Solvent usage by workflow step, conditional on step presence. Cell values are percentages.

The solvent head uses the asymmetric loss of Ridnik et al. (2021),

$$\begin{aligned}\mathcal{L}_{\text{sol}} = -\sum_{t,k} \big[& s_{t,k}(1-p_{t,k})^{\gamma_+} \log p_{t,k} \\ & + (1-s_{t,k})(p^-_{t,k})^{\gamma_-} \log(1-p^-_{t,k})\big], \\ p^-_{t,k} &= \max(p_{t,k}-\delta, 0),\end{aligned} \tag{A2}$$

with $\gamma_+ = 0$, $\gamma_- = 4$ and $\delta = 0.05$. This reduces contributions from easy negatives in a sparse 5×378 grid. The mean of 8.25 distinct solvent tokens collapses over steps and is not the number

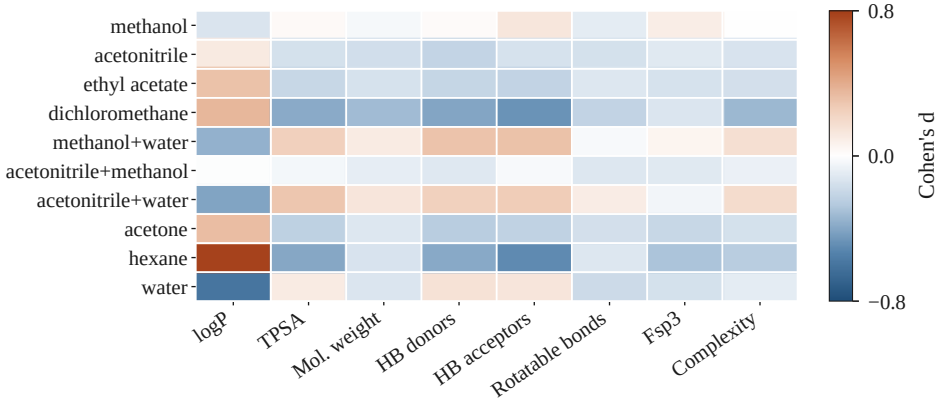


Figure A5: Descriptor associations with elution solvents. Values are standardised descriptor differences between pollutants that do and do not receive a solvent.

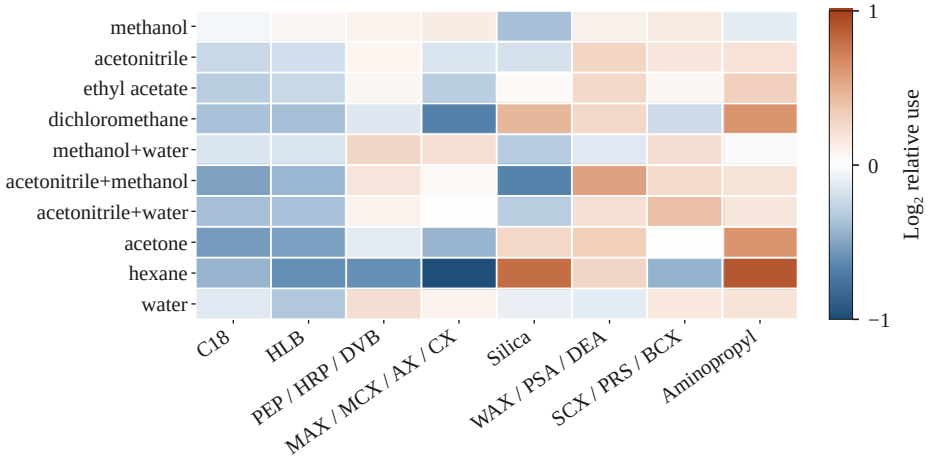


Figure A6: Cartridge associations with elution solvents. Cartridge preference is expressed as $\log_2[P(\text{solvent} \mid \text{cartridge})/\text{row mean}]$.

of positive step–solvent cells. Step prediction uses positive-weighted binary cross-entropy. Ratio prediction uses masked cross-entropy over 20 equal-width buckets with label smoothing 0.1.

Auxiliary losses. Let P contain the positive labels of a sample and let N be the non-positive candidate labels used in the denominator, so $P \cap N = \varnothing$. The set-ranking term is

$$\mathcal{L}_{\text{rank}} = -\frac{1}{|P|} \sum_{j \in P} \log \frac{\exp(u_j)}{\sum_{\ell \in P \cup N} \exp(u_\ell)}. \tag{A3}$$

A neighbourhood term connects each sample to its eight nearest within-batch neighbours in cosine descriptor space,

$$\mathcal{L}_{\text{sm}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{N}_8(i)} w_{ij} \|\mathbf{z}_i - \mathbf{z}_j\|_2^2, \tag{A4}$$

The reported weight is $w_{ij} = \cos(\widetilde{\mathbf{x}}_i, \widetilde{\mathbf{x}}_j)$. Rank-Gauss centring permits negative cosine similarities, and selecting nearest neighbours alone does not prove that all selected weights are non-negative. Negative terms favour separation; the expression is therefore not guaranteed to be an attractive, non-negative smoothing penalty. A safeguarded alternative would use

$$w_{ij}^{+} = \max\{\cos(\widetilde{\mathbf{x}}_i, \widetilde{\mathbf{x}}_j), 0\}. \tag{A5}$$

This is a proposed safeguard, not a claim about the implementation that produced the tables. Its effect requires checking the weight construction and, if it changes, rerunning training. The reported step and

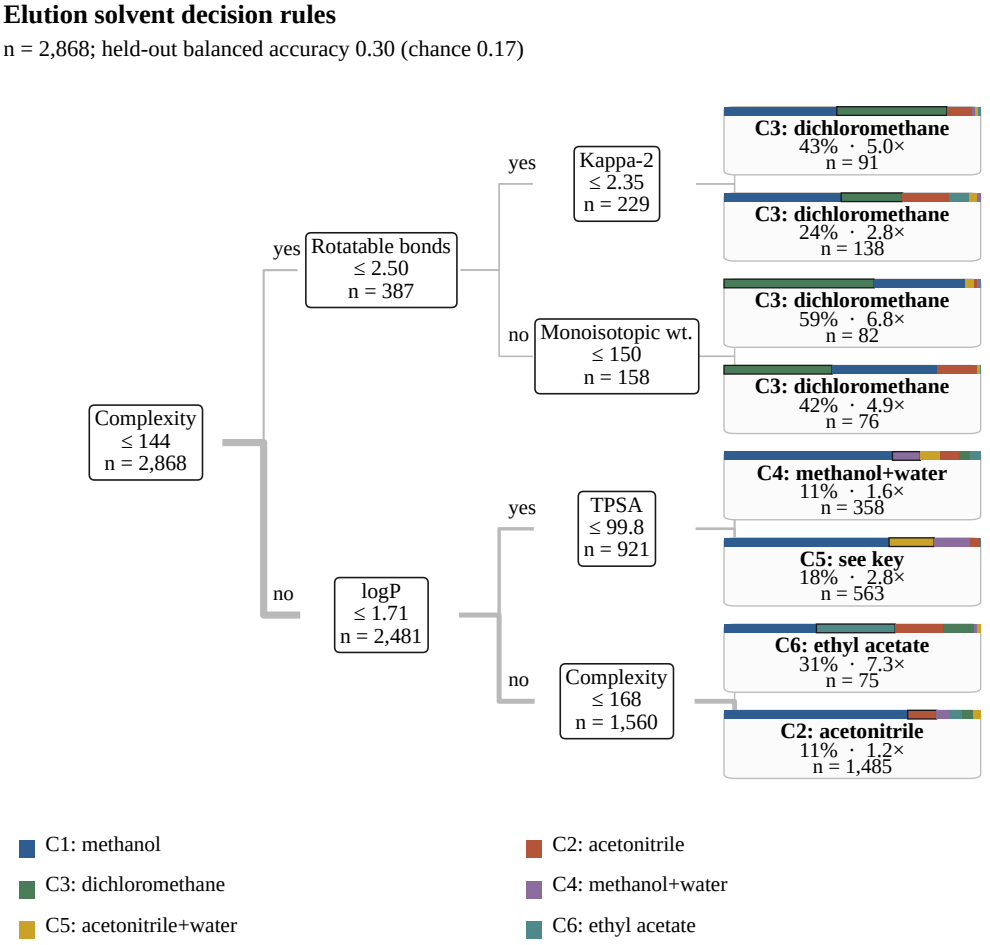


Figure A7: Descriptor-level decision rules for the elution solvent, using the same construction as Figure 6; C1–C6 identify the solvents in this figure's key.

ratio task weights are one and the neighbour weight is 0.01; solvent and cartridge-ranking weights distinguish operating points. The choice between all non-positive labels and sampled negatives in N also remains an implementation detail that must be specified.

Conditional head and auxiliary variant. Writing $\mathbf{e}_t \in \mathbb{R}^{d_e}$ and $\mathbf{f}_k \in \mathbb{R}^{d_f}$ makes the required dimensions explicit: g maps $\mathbb{R}^{512+d_e+d_f}$ to $\mathbb{R}^{20}$ before softmax. These symbols do not fix the embedding widths or the internal layers of g. SPEGen-EM adds a set-size predictor beyond the four core heads and uses a validation-tuned output quantile. Its parameterisation, supervision loss, loss coefficient and mapping from predicted size to the output set are not specified by the core architecture equations. They are required for an independent implementation; they cannot be inferred from its average predicted set size of 1.4.

Optimisation, decoding and ensemble. We use the AdamW implementation (Loshchilov & Hutter, 2019) with learning rate 10^{-3}, weight decay zero, batch size 64, gradient clipping at 1.0 and 60 epochs. With zero decay, its decoupled weight-decay term is inactive. Checkpoints are selected by a validation score that combines the end-to-end metrics defined below. For each label with at least five positive validation examples, a threshold is selected over 41 grid points; rarer labels share a global shift. SPEGen-EM additionally predicts set size and tunes an output quantile. SPEGen-Bag contains eight SPEGen-Chain members trained on independent bootstrap resamples (Breiman, 1996); member probabilities are averaged before thresholding, and validation and test splits are never resampled.

Table A2: Implementation details of SPEGen

Item	Value
Corpus split	11,441 pollutants; 9,152 train, 1,144 validation, 1,145 test (random, seed 42)
Input	21 descriptors; rank-Gauss transform fitted only on the training split
Encoder	MLP with two 512-unit hidden layers, ReLU and dropout 0.3
Core heads	42-way cartridge, 5-way step, 5×378 solvent and 20-bucket ratio; SPEGen-EM additionally uses a set-size predictor
Classification losses	Cartridge focal ($\gamma = 1.5$), weighted step BCE and solvent asymmetric loss ($\gamma_- = 4$, clip 0.05)
Auxiliary losses	Set-level ranking; top-8 within-batch neighbour smoothing with weight 0.01
SPEGen-Bal weights	Cartridge rank weight 0.5; solvent task weight 1.0; micro-F1 threshold tuning; no set-size head
SPEGen-Chain weights	Cartridge 1.0, step 1.0, solvent 0.1, ratio 1.0; cartridge rank weight 2.0
Optimiser	AdamW, learning rate 0.001, weight decay 0, batch 64, 60 epochs, gradient clip 1.0
Decoding	Per-label thresholds and a global shift tuned on validation; solvents gated by predicted steps
SPEGen-EM decoding	Additional set-size predictor; EM/Jaccard objective and validation-tuned output quantile
SPEGen-Bag ensemble	Eight independent SPEGen-Chain bootstrap members; probabilities averaged before micro-F1 decoding
Repeats	Four initialisation seeds per single-network configuration; tables report mean ± s.d.

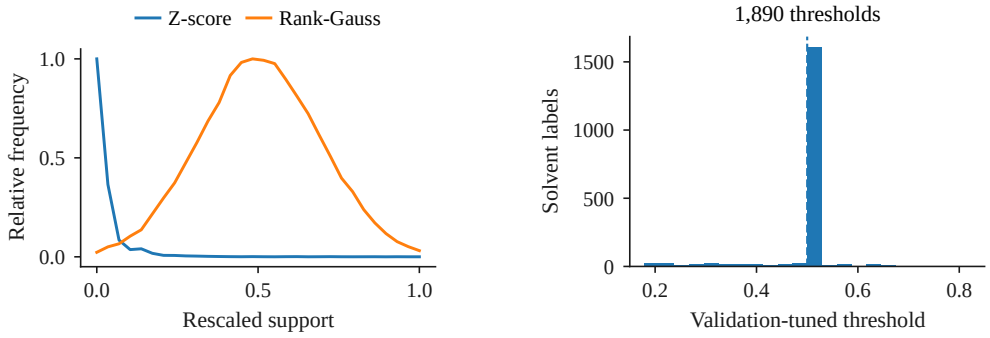


Figure A8: Input and decoder distributions. (a) The descriptor with the largest excess kurtosis before and after rank-Gauss transformation, using the central 0.3–99.7 percentile interval and independently rescaled supports. (b) Validation-tuned thresholds for all 1,890 step–solvent labels; the dashed line is 0.5.

D Evaluation details

Comparison protocol and reporting. All comparators use the same descriptor information and pollutant split. Multi-label subtasks use one-vs-rest wrappers where needed; their outputs are assembled into the same component representation for scoring. Shared inputs and scoring do not establish equal hyperparameter-search, threshold-calibration or computation budgets. Outer-loop bagging of the strongest baselines and repeated baseline fits are not reported. Random forests and boosting aggregate trees internally, but are not matched to the eight-network ensemble used by SPEGen-Bag.

C-F1 and S-F1 denote cartridge and solvent micro-F1; C-EM denotes cartridge exact match and C-top1 cartridge top-1 accuracy. SSS, HC and GUS denote scheme similarity, hierarchical consistency and global utility. Annotation-dependent denominators can be smaller than the full test set. Table entries use decimals, except GUS on its reported 100-point scale; figure axes marked % rescale the corresponding decimals by 100. Single-network entries report mean ± standard deviation over four initialisations. Baselines and the eight-member ensemble are single reported evaluations. Bold

identifies the best reported column value, not statistical significance; ratio MAE is minimised and other performance scores are maximised. This is one random pollutant split, not scaffold-disjoint, source-disjoint or external validation.

Task-level measurements. For a labelled set Y_i and predicted set $\widehat{Y}_i$, let TP, FP and FN be counts pooled over the eligible samples and label indices. The standard set measurements are

$$F_{1,\text{micro}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}, \qquad \text{EM} = \frac{1}{n}\sum_i \mathbb{1}[\widehat{Y}_i = Y_i]. \tag{A6}$$

With $a^r_{i,t,k} = 1$ for a reported numeric ratio and zero otherwise, the corresponding masked ratio summaries are

$$\text{MAE}_r = \frac{\sum_{i,t,k} a^r_{i,t,k} |\widehat{r}_{i,t,k} - r_{i,t,k}|}{\sum_{i,t,k} a^r_{i,t,k}}, \qquad A_r(\epsilon) = \frac{\sum_{i,t,k} a^r_{i,t,k} \mathbb{1}[|\widehat{r}_{i,t,k} - r_{i,t,k}| \le \epsilon]}{\sum_{i,t,k} a^r_{i,t,k}}. \tag{A7}$$

Missing numeric entries are not zero-valued targets. Ratio accuracy is measured against the aggregated ratio, not against successful extraction in a laboratory.

Top-ranked selections and annotation masks. Let (j^*_i, t^*_i, k^*_i) denote the evaluator's selected cartridge, step and solvent. The described component-membership chain event is

$$E_i = c_{i,j^*_i}\, m_{i,t^*_i}\, s_{i,t^*_i,k^*_i}. \tag{A8}$$

Chain accuracy is averaged over pollutants with protocol annotation. It tests one selected triple, not an exact five-step procedure, and membership in union targets does not establish that the triple appeared together in a source method. Ratio-verified Hit@1 additionally requires $a^r_{i,t^*_i,k^*_i} = 1$ and $|\widehat{r}_{i,t^*_i,k^*_i} - r_{i,t^*_i,k^*_i}| \le 0.10$. A missing ratio cannot certify a hit. The precise selection rule, tie handling and Hit@1 denominator must accompany the evaluator; excluding a missing selected pair from a denominator and counting it as a miss are different estimands.

On a common eligible population A, conditional decomposition satisfies

$$P(C, M, S \mid A) = P(C \mid A)P(M \mid C, A)P(S \mid C, M, A). \tag{A9}$$

An overall cartridge score and conditional scores from an annotation-filtered subset need not obey this product. Thus the diagnostics in Figure 4(b) motivate checking solvent ranking, but do not establish an exact numerical factorisation without common masks and counts. Unannotated pollutants cannot count as chain hits solely because their cartridge is correct. Earlier omission of annotation checks can inflate results; removing the omission does not inflate them.

Composite scores and consistency. SSS is described as a weighted combination of component Jaccard scores and ratio accuracy, and GUS as a weighted combination of task-level F1 and ratio tolerance. The numerical weights, sample-versus-task aggregation and missing-component renormalisation are not specified in the experiment description. Consequently, the retained SSS and GUS values cannot yet be independently reproduced from these definitions and are treated as auxiliary, not as the sole evidence for improvement.

For consistency, define $e_{i,t} = \mathbb{1}[\sum_k \widehat{s}_{i,t,k} > 0]$. The two distinct conditions are

$$e_{i,t} \le \widehat{m}_{i,t} \quad \text{(no solvents for absent steps)}, \qquad e_{i,t} = \widehat{m}_{i,t} \quad \text{(also no empty present steps)}. \tag{A10}$$

Equation 6 guarantees the first, not the second. A score checking only the first condition on post-gating outputs would be one. HC values below one therefore require another condition or a different evaluation stage; the published scalar values alone do not identify which. The exact HC implementation remains to be checked. Neither condition verifies sorbent compatibility, feasible mixture proportions or analyte recovery.

Strict end-to-end exact recovery was reported below 0.05 for every model. This low level is a substantive limitation of exact recovery, not evidence that a bench-ready protocol has been obtained; model-specific values are not available in the reported tables.

Statistical treatment and descriptive analyses. Each single-network configuration is repeated over four initialisation seeds on the fixed split. The reported ± 2 s.d. band of 2.4 percentage points is a heuristic scale for single-run comparisons, not a confidence interval for paired model differences. Four initialisations do not measure uncertainty over data splits, and one ensemble evaluation does not measure uncertainty over bootstrap ensembles. Some identical means in Tables 2 and 3 have different reported standard deviations; their variance conventions must be reconciled from the per-seed records before inferential use. Descriptive enrichment analyses use the full corpus because they characterise the data and do not affect model selection. Descriptor axes are quantile-binned, cells with fewer than 40 molecules are blank, binomial proportions use Wilson intervals (Wilson, 1927), and descriptor contrasts use Cohen's d (Cohen, 1988). Shallow trees use balanced class weights and a held-out 20% partition.

Software. Data construction, modelling, training, decoding and evaluation are implemented in Python and PyTorch (Paszke et al., 2019). Baselines use scikit-learn (Pedregosa et al., 2011), XGBoost and LightGBM.

E Additional quantitative analyses

Scope of ablation. The stage-wise table follows a cumulative development path. A leave-one-out or factorial design would be needed to separate effects from interactions. Bootstrap members see about 63% of distinct training examples, and the reported comparison associates aggregation with gains without separating training-sample diversity from increased inference capacity. Initialisation-only ensembles were reported to give small changes, not a compute-matched attribution to resampling alone.

Figure A9 gives the precision–recall and predicted-set-size comparisons discussed in Section 5.5. They describe alternative decoding objectives, not additional performance on a new data split.

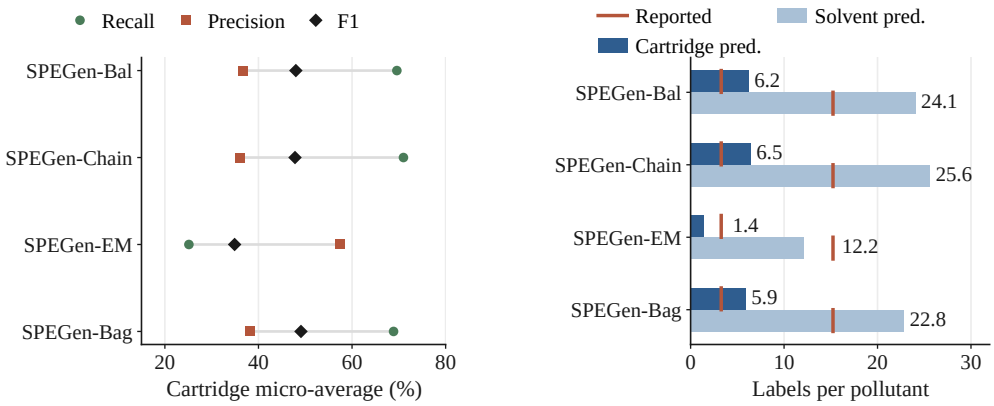


Figure A9: SPEGen decoder trade-offs. (a) Cartridge precision, recall and F1 for the four operating points. (b) Predicted and reported cartridge and solvent set sizes. F1-oriented variants overshoot cardinality, whereas exact-match decoding undershoots it.

The complete comparison of six baselines and four SPEGen operating points is reported in Tables 1 and 2. The additional interventions retain their reported single-run scores, but their control identities are unresolved. The oracle-size analysis keeps candidate ranking fixed and is interpreted only as a decoding diagnostic.

For Table A3, the implied control is reconstructed arithmetically as $c = a - \Delta/100$ from a reported score a and change Δ in percentage points. It is not a verified control run. Rounded contrasts imply several control levels, inconsistent with assigning the first three rows to one pre-transform control; configuration-specific controls must be verified before comparative conclusions are drawn.

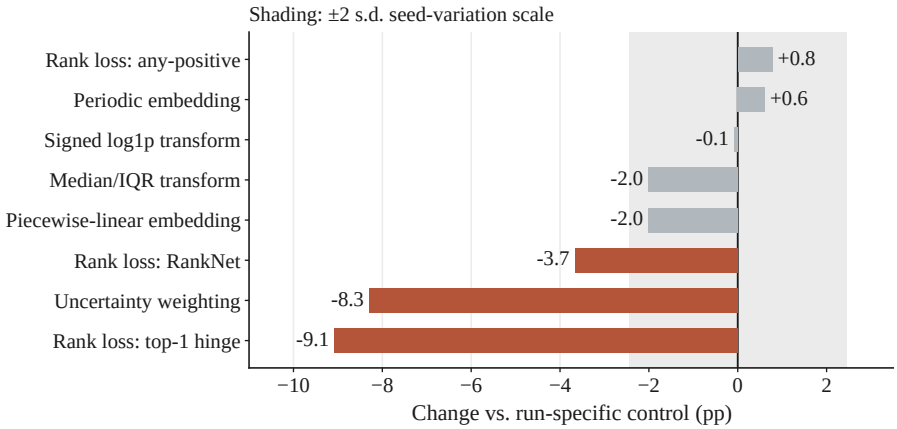


Figure A10: Reported single-run intervention contrasts. Control identities are unresolved (Table A3); the shaded 2.4-percentage-point band is a heuristic seed-variation scale, not a significance test.

Table A3: Additional interventions: reported results and arithmetic control check

Intervention	**C-top1**	**Reported Δ (pp)**	**Implied control**
Rank loss: top-1 hinge	0.4961	−9.1	≈ 0.5871
Uncertainty weighting	0.4262	−8.3	≈ 0.5092
Rank loss: RankNet	0.5502	−3.7	≈ 0.5872
Piecewise-linear embedding	0.4891	−2.0	≈ 0.5091
Median/IQR transform	0.5668	−2.0	≈ 0.5868
Signed log1p transform	0.5860	−0.1	≈ 0.5870
Periodic embedding	0.5153	+0.6	≈ 0.5093
Rank loss: any-positive	0.5948	+0.8	≈ 0.5868

The oracle-size diagnostic in Table A4 uses one SPEGen-Chain run, not the four-seed mean of the main comparison. Score rankings are held fixed while set sizes are replaced. SSS and GUS retain the definition limitations discussed in Appendix D. A ranking intervention would additionally require verification of the score arrays consumed by the chain evaluator.

Table A4: Fixed-ranking oracle-size diagnostic (one SPEGen-Chain run)

Metric	**As decoded**	**Oracle set size**
Cartridge F1	0.4805	**0.6532**
Cartridge exact match	0.0183	**0.2498**
Solvent F1	0.4564	**0.6470**
Chain accuracy	**0.4909**	**0.4909**
Scheme similarity	0.5085	**0.6286**
Global utility	61.35	**72.34**
Predicted cartridge set size	6.43	3.27

Knowing the target set size can improve recovery under a fixed ranking, as in Table A4, but is not a universal F1 upper bound: under an imperfect ordering, selecting more than the true number of labels can sometimes give higher F1. Nor do low R^2 values for one count predictor establish that all models using the same descriptors must fail. Invariance of chain accuracy to a set-size change follows from preserving the top-ranked scores, not from a proof of independence between cardinality and ranking errors. A perfect-ranking oracle would need to change the scores consumed by top-1 evaluation; replacing decoded set indicators alone would not constitute such an intervention.

F INTERPRETATION OF AGGREGATED SUPERVISION

Marginal alternatives are not joint procedures. Suppose two reported methods pair cartridge c_1 with solvent system s_1 and cartridge c_2 with system s_2. The aggregate target contains $\{c_1, c_2\}$ and $\{s_1, s_2\}$, but provides no evidence for the cross-pair (c_1, s_2). Scoring its components as positive therefore does not establish a compatible method. The same loss of co-occurrence applies across steps and to choices recorded under different matrices or concentrations. The shared encoder can learn descriptor associations, but it does not restore the source-method linkage discarded by aggregation. Top-1 selection and step gating alone cannot turn these marginal sets into a validated execution sequence.

Ratio averages and mixture constraints. Both target construction and expectation decoding can interpolate between reported modes. For example, fractions 0.5 and 0.9 have arithmetic mean 0.7, corresponding to 7:3 in a binary mixture. The mean is within the unit interval but need not have appeared in any source method. This is an illustrative arithmetic example, not a measured frequency of conflicts in this corpus. The present tables evaluate prediction of aggregated fractions; they do not establish preservation of conventional mixtures. Argmax bucket decoding, $\widehat{r}_{\mathrm{mode}} = \nu_{\arg\max_b q_b}$, would avoid averaging predicted bucket modes but would not reverse averaging already performed in the labels or guarantee that a bucket centre is a conventional ratio. Method-level multimodal targets and alternative decoders require a separate experiment.

A solvent token can represent a neat solvent or a composite system such as methanol+water. Multiple tokens within a union are alternatives, not necessarily simultaneous co-solvents. Imposing a simplex across all positive union tokens would therefore be inappropriate. A physically interpretable composition model would first identify one method and its component list, then enforce non-negativity and a sum-to-one constraint within that mixture. The current independent bounded ratio outputs do not impose this constraint and do not certify a complete mixture.

Positive-only reporting. The classification objectives use absent archive labels as computational negatives, not as experimentally verified failures. Even under the idealised assumption that every reported choice was feasible in its recorded context, reporting and feasibility remain different events. Let V indicate feasibility, D indicate selection and documentation, and $O = VD$ indicate an observed positive. Conditional on molecular descriptors x and analytical context a,

$$P(O = 1 \mid x, a) = P(D = 1 \mid V = 1, x, a)P(V = 1 \mid x, a). \tag{A11}$$

This factorisation requires no independence assumption. Without knowledge of the documentation process, the feasibility probability is not identified by the observed-positive rate; omitting context adds another source of ambiguity. Focal and asymmetric losses address imbalance but do not estimate that documentation process. Likewise, decoder tuning and bagging do not by themselves calibrate out-of-distribution confidence. Reported scores measure recovery of recorded choices, not probabilities of successful extraction. Negative experiments, contextual supervision and prospective assessment would be needed for stronger claims.

G SCOPE OF COMPARISON AND REPRODUCIBILITY LIMITS

Table A5: Implementation details not recoverable from the reported summaries

Component	Additional information needed for independent reproduction
Input transform	Interpolation rule, clipping/extrapolation at training extrema, missing-value handling and exact descriptor list.
Ratio and size heads	Embedding widths, internal layers of g, bucket boundary handling, and the auxiliary size-head architecture, loss and quantile rule.
Training objective	Candidate-negative selection, cosine-weight sign handling, and loss masking for pollutants without protocol annotations.
Model selection	Exact checkpoint score, threshold grid limits/global-shift rule, and per-model hyperparameter/decoder search budgets.
Evaluation	SSS/GUS weights and reductions; HC conditions and pre/post-gating stage; all eligibility masks, denominators and tie rules.
Run provenance	Per-seed outputs and standard-deviation convention, intervention-specific control IDs, and score-array changes in oracle runs.

The random pollutant split separates identifiers within one corpus, not necessarily scaffolds or source publications. Possible structural overlap limits extrapolation claims; it does not prove test-label leakage. Scaffold- and source-grouped evaluation, modern tabular models, dependency-aware baselines and outer-loop bagged tree baselines are not reported experiments. Public reaction resources such as the Open Reaction Database (Kearnes et al., 2021) could support a separately defined transfer task, but are not automatically interchangeable with these SPE targets.

Equal descriptors and a common evaluator do not establish matched comparisons: preprocessing, label-dependency treatment, search space and budget, checkpoint selection, threshold calibration and fitting/inference cost must also be recorded. A shared encoder does not by itself demonstrate an advantage over explicit dependency models, and sequential ablation measures conditional increments rather than isolated component effects. Table A5 identifies unresolved implementation details; no guessed choices or modified decoders, neighbour weights or splits are treated as having produced the existing results.